

Calculations				
	Sheep	Pig	Cow	Goat
Meat Diet Portion	1.53%	1.45%	20.35%	2.47%
Meat MCal Target	27,874.32	26,508.86	371,359.40	45,107.43
Potential Dairy Value attached to each juvenile	132.54	0.00	1015.20	126.90
Value per juvenile	48.00	160.00	640.00	48.00
Adjusted Dairy Value attached to juvenile based on Dairy Utilization	66.27	0.00	507.60	63.45
Culled Mothers Value	14.77	4.41	105.26	14.77
Culled Adult Males	0.74	0.22	5.26	0.74
Final Value per juvenile	63.51	164.63	750.53	63.51
Value MCals	63.51	164.63	750.53	63.51
Offspring required Year ⁻¹	438.91	161.02	494.80	710.27
MCals from meat	27874.32	26508.86	371359.40	45107.43
Max Utilized Mcals allowed from Dairy	29086.73	0.00	251159.79	45066.45
MCals actually Utilized from Dairy	29086.73	0.00	251159.79	45066.45
Percent of diet from their meat	1.53%	1.45%	20.35%	2.47%
Percent of Diet from their Dairy	1.59%	0.00%	13.76%	2.47%
Kg Fleece	3,317.93	0.00	0.00	0.00
1 Year+ (age) skins	163.38	6.42	95.16	264.21
Yearly Offspring skins	799.11	190.30	729.04	1,292.78
Birthing Events per year	1.00	1.86	1.00	1.00
Offspring Mother ⁻¹ Year ⁻¹	0.56	5.48	0.68	0.56
Mothers Step 1	783.77	29.41	727.65	1,268.33
Males Step 1	219.46	80.51	247.40	355.13
Females Step 1	219.46	80.51	247.40	355.13
Replacement Mothers per year	157.05	6.18	91.14	253.97
Breeding Males Required	41.15	1.54	38.20	66.59
Additional Mothers	643.20	5.35	344.47	1,040.20
Males Step 2	180.10	14.64	117.12	291.26
Females 2	180.10	14.64	117.12	291.26
Total Mothers	940.83	35.59	818.79	1,522.30
Total Males	399.55	95.15	364.52	646.39
Total Females	39.36	65.88	130.28	63.88
Total Offspring	799.11	190.30	729.04	1,292.78
Total Animals	1,420.89	198.16	1,351.79	2,299.16
Units per Hectare	1.00			
Total Adults per Hectare	1.71	0.26	0.42	1.71
Total Offspring per Hectare	0.84	1.15	0.25	0.84
	1.05%		12.78%	1.70%
Total Adults	1,625	42	1,201	2,629
Total Offspring	799	190	729	1,293
Days for Maintenance	5.00	220.00	0.00	5.00
Dry Mothers	141.72	0.00	89.75	229.52
Other	12,785.72	0.00	0.00	20,684.46
MCals Dry Mothers	165,528.43	0.00	189,999.54	268,081.89
total Maintenance MCals	178,314.14	0.00	189,999.54	288,766.36

MCals Required				
	Sheep	Pig	Cow	Goat
Gestating	475,868	86,880	1,220,411	769,850
Lactating	907,866	30,885	1,312,270	1,468,726
Maintenance	178,314	0	190,000	288,766
Adult Males	48,061	0	80,872	77,774
Offspring	219,355	191,248	2,668,282	354,868
TOTAL	1,829,464.1	309,012.6	5,471,834.3	2,959,984.5

Animal Diet Breakdown					
	Sheep	Pig	Cow	Goat	
	Crop Barley	Barley	Barley	Barley	
Grams ⁻¹ Straw/Chaff	1000.0	0.0	4956.0	1000.0	
MCal kg ⁻¹					
Grams ⁻¹ Grain	750.0	750.0	1500.0	750.0	
MCal kg ⁻¹					
Days	90.0	90.0	90.0	90.0	
Number of Animals	799.1	190.3	729.0	1,292.8	
MCals from Straw/Chaff	102125.9	0.0	461756.2	165217.1	
MCals from Grain	161819.2	38535.0	295260.7	261787.7	
MCals from other feed	263945.1	38535.0	757016.9	427004.9	

Diet Calculations		
Plants Mcal	952,187.030	
Meat Mcal	547,500	
Dairy	325,312.97047643	
	Perfect	
Plants %	52.17%	
Meat %	30.00%	
Dairy %	17.83%	
check	Perfect	
Percent		
Domestic Meat	25.80%	
Dairy	17.83%	
Wild Meat	4.20%	
Crops	46.96%	
Wild Plants	5.22%	
check	Perfect	
new dairy	325312.97	
potential dairy < limit	TRUE	
new limit	70.0%	
the limit greater than that left to satisfy	TRUE	
limit	1	
limit dairy checked	FALSE	
Mcalories		Surplus
Domestic Meat	470,850.000	
Dairy	325,312.970	
Wild Meat	76650.000	
Crops	856,968.327	
Wild Plants	95,218.703	
check	Perfect	
SURPLUS DAIRY	0.000	
This could account for additional % of diet	0.00%	

Herd Modelling

	Sheep	Pig	Cow	Goat				
Mothers Req'd Males for Slaughter Females for Slaughter	941	36	819	1522				
	400	95	365	646				
	39	66	130	64				
Offspring available for Consumption	799	190	729	1293				
Gestation Req'd	261372	73516	828293	422964				
Lactation Req'd	498648	26134	890638	806935				
Maintenance Req'd	593258	0	243932	959820				
Adult Males	48061	0	80872	77774				
Offspring Required	262507	208374	3193190	424678				
Energy Required	1663847	308024	5236924	2692171				
Ewe Fleece Kg Yr-1	2117							
Ram Fleece Kg year ⁻¹	82							
Juvenile Fleece Kg Yr-1	1119							
Kg Fleece Year-1	3318	0	0	0				